NATURAL RESOURCE INVESTMENT RISK, RETURN, AND DIVERSIFICATION

Learning Outcome

- analyze sources of risk, return, and diversification among natural resource investments

Commodity prices have risk and return drivers that are often related to but do not always directly coincide with the timing of the economic cycle affecting the prices of common equity and debt securities. Commodities, farmland, and timberland have different return drivers and cycles. Commodities are priced on a second-by-second basis on public exchanges, whereas land generally has an infrequent pricing mechanism and may include imprecise estimates, as opposed to actual transactions. Keeping these market structure differences in mind helps investors consider their relative benefits and challenges.

Commodities

Physical commodity supply dynamics are determined by production (in the case of hard commodities), seasonal crop yields (for soft commodities), and inventory levels in the short term, while commodity end user/consumer use of these basic inputs drives ultimate demand. Supplies of physical commodities are determined by production and inventory levels and secondarily by the actions of non-hedging investors. Demand for commodities is determined by the needs of end users and secondarily by the actions of non-hedging investors. Investor actions can both dampen and stimulate commodity price movements, at least in the short term. Gold, a physical commodity and a precious metal, is often a preferred safe haven given its historical use as a store of value among investors and as a non-currency based reserve among central banks.

Producers cannot alter commodity supply levels quickly because extended lead times are often needed to affect production levels. For example, agricultural output may be altered by planting more crops and changing farming techniques, but at least one growing cycle must pass before there are results. And at least one factor beyond the producer's control —the weather—will significantly affect output. Building the necessary infrastructure for increased oil and mining production may take many years, involving both developing the mine itself and the necessary transportation and smelting components. For commodities, suppliers' inability to quickly respond to changes in demand may result in supply too low in times of economic growth and too high when the economy slows. And despite advancing technology, the cost of new supply may grow over time.

Investing in commodities is motivated by its potential for returns, portfolio diversification, and inflation protection. Investors may choose commodities if they believe prices will increase in the short or intermediate term. Commodity futures contracts may offer investors a liquidity premium or other trading opportunities, creating the prospect for a positive real return. In the 30-year period referenced in Exhibit 6, commodity investments outperformed global stocks and global bonds but with much higher volatility.

Exhibit 6:

Historical Returns of Commodities, Q3 1992–Q2 2022 (quarterly data)

	Global Stocks	Global Bonds	Commodities
Annualized return			
1992:Q3–2022:Q2	6.89%	4.39%	7.81%

1992:Q3-2022:Q2	16.76%	6.14%	24.39%
Worst calendar year	−43.54%	−5.17%	−42.80%
	(2008)	(1999)	(2008)
Best calendar year	31.62%	19.66%	50.30%
	(2003)	(1995)	(2009)

Sources: Global stocks, MSCI ACWI; global bonds, Bloomberg Barclays Global Aggregate Index; commodities, S&P GSCI Total Return.

Exhibit 6 shows a summary of investment performance and volatility of global stocks, global bonds, and commodities over a 30-year time horizon from Q3 1992 through Q2 2022. Commodities exhibit the highest average return and the highest volatility among the three asset classes. The worst performance for both global stocks and commodities occurred during 2008, coinciding with the middle of the global financial crisis of 2007–2009. As global economies began recovering in 2009, commodities exhibited their best calendar-year performance. While the data may imply that commodities behave similarly to global stocks, we directly address the correlation between these two asset classes later in this lesson. Overall demand levels are influenced by global manufacturing dynamics and economic growth. When demand levels and investors' orders to buy and sell during a given period change quickly, the resulting mismatch of supply and demand may lead to price volatility.

Farmland and Timberland

Farmland and timberland, in contrast, are far less frequently traded and derive their value from different sources. In the case of farmland, it is multiple growing seasons over time that generate the return. In the case of timberland, it is the longer forest/tree growth cycle and the demand for lumber that determine returns once the lumber has been cut down. The size of the global investable farmland market is estimated at approximately USD1 trillion, with a relatively small proportion, less than 5%, held by institutional investors. Institutional investors hold about one-quarter of the global investable timberland base, valued at roughly USD285 billion. A large majority of timberland held by these investors is located in the United States, Australia, and New Zealand.

Turning to land, Exhibit 7 provides a comparison of returns on US timber and farmland. The National Council of Real Estate Investment Fiduciaries (NCREIF) constructs a variety of appraisal-based indexes for property, timberland, and farmland. Over the 30-year time period from Q3 1992 to Q2 2022, farmland had the higher annualized return and timber had the higher standard deviation.

Exhibit 7:

Historical Returns of US Real Estate Indexes, Q3 1992–Q2 2022 (quarterly data)

	NCREIF Data	
	Timberland	Farmland
Annualized return	8.69%	10.95%
Annualized standard deviation	6.76%	5.88%
Worst calendar year	−5.30%	2.02%
	(2001)	(2001)
Best calendar year	22.36%	33.90%
	(1993)	(2005)

Although the data in Exhibit 7 make farmland appear to be a very attractive investment, it has definite risks. Liquidity is very low, the risk of negative cash flow is high because fixed costs are relatively high (land requires care and crops need fertilizer, seed, and so on), and revenue is highly variable based on weather. The risks of timberland and farmland are

commercial and residential real estate properties. Drought and flooding can dramatically decrease the harvest yields for crops and thus the expected income stream.

In contrast to the local nature of real estate, farmland and timberland are exposed to more global risks given that these investments generate commodities that are globally traded and consumed. For example, there have been interruptions in world trade, and growing agricultural competition has resulted in declining grain prices. Therefore, it seems difficult to repeat these returns over the next 30 years. Timberland and farmland investments should consider the international context as a major risk factor.

Finally, investment in vacant or raw land that has not been developed or prepared for construction generally involves greater risk than farmland or timberland.

Inflation Hedging and Diversification Benefits of Natural Resource Investments

Investors often consider commodity investments as a hedge against inflation and as a source of portfolio diversification relative to a portfolio of traditional assets (i.e., stocks and bonds).

Hedge against Inflation

The argument for commodities as a hedge against inflation derives from some commodity prices being components of inflation calculations. Commodities, especially energy and food, affect consumers' cost of living. The volatility of commodity prices, especially energy and food, is much higher than that of reported consumer inflation. Consumer inflation is computed from many products, including housing, whose prices change more slowly than commodity prices, and inflation calculations use statistical smoothing techniques and behavioral assumptions. Exhibit 8 shows a summary of calendar year returns on global stocks, global bonds, commodities, farmland, and timberland segmented by whether the US CPI (i.e., inflation) is above or below its median of 2.26%.

Exhibit 8:

Historical Asset Class Returns Divided by Median US CPI, 1993–2021 (annual returns)

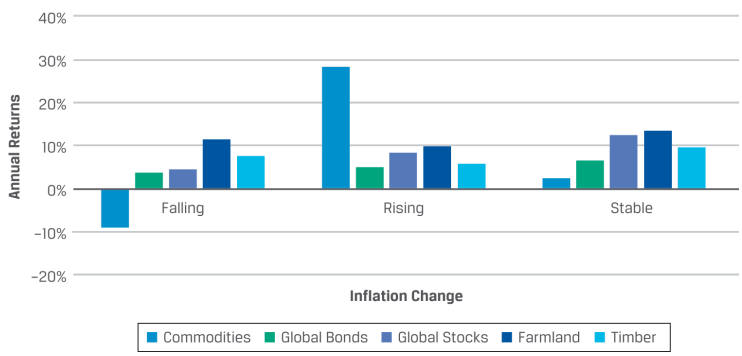
	Global Stocks	Global Bonds	Commodities	Farmland	Timberland
Higher inflation	+9.40%	+5.66%	+22.87%	12.78%	10.44%
Lower inflation	+5.43%	+4.18%	−9.26%	9.85%	5.70%

Sources: Global stocks, MSCI ACWI; global bonds, Bloomberg Barclays Global Aggregate Index; commodities, S&P GSCI Total Return; Farmland and Timberland, NCREIF

The data from Exhibit 8 covering 29 full calendar years are consistent with the idea that investments in commodities perform well when inflation is higher and perform poorly when inflation is lower. This difference also holds if the 29 calendar years are split into low, middle, and high inflation rates. In fact, the performance distinctions on commodities are even larger using the three inflation categories instead of only two. However, there is less evidence that investment performance of farmland or timberland differs significantly in differing inflation environments.

As an alternative to presenting asset class returns during higher or lower inflation, Exhibit 9 shows average return data for commodities, global bonds, and global stocks when inflation has moved lower, higher, or very little. Here, stable inflation is defined as a less than a 10 bp move from year to year, so falling and rising inflation regimes are those years with larger year-to-year changes in inflation.

Exhibit 9:



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Sources: Global stocks, MSCI ACWI; global bonds, Bloomberg Barclays Global Aggregate Index; commodities, S&P GSCI Total Return Farmland and Timber, NCREIF

Consistent with the data shown in Exhibit 8, the returns presented in Exhibit 9 highlight that commodities provide greater returns in rising inflation environments at a cost of negative returns as inflation rates decline. In stable inflation environments, commodity returns have tended to be low but positive. None of the other asset classes, including farmland and timberland, demonstrate such large contrasts in performance based on changing inflation.

Portfolio Diversification

Farmland, timberland, and commodities exhibit potential for portfolio diversification. Historically, all three of these asset classes have low correlations with investment returns from traditional assets (i.e., stocks and bonds) during the business cycle. Exhibit 10 shows the quarterly correlation between farmland, timberland, selected commodities, global equity, and global bond indexes from Q3 1992 through Q2 2022. All three of the alternative asset classes show correlations of approximately zero with global bonds. Timberland shows the lowest correlation of the alternative asset classes with global stocks, at approximately 0.02. Farmland's correlation with global stocks is still low, at approximately 0.12, while commodities demonstrate the least diversification potential with global stocks, given the correlation of 0.41. The correlations of stocks, bonds, and commodities are expected to be positive because all the assets have some exposure to the global business cycle. The commodity index, the S&P GSCI (Goldman Sachs Commodity Index), is heavily weighted toward the energy sector, with each underlying commodity possibly exhibiting unique behavior.

The data in Exhibit 10 also complement the earlier discussion of the potential for asset classes to serve as inflation hedges. The correlation between commodities and inflation of 0.54 is much higher than that of other asset classes with inflation and is consistent with the idea of commodities as an inflation hedge. Neither farmland (−0.17) nor timberland (0.02) shows a high degree of correlation with inflation, implying that these asset classes are less effective as inflation hedges. However, the prior section suggests that the lack of correlation has more to do with their investment performance not suffering during periods of low or declining inflation. Commodity returns, in contrast, are negative during low or declining inflation, as shown earlier.

Exhibit 10:

Historical Commodity Return Correlations, 1990–Q1 2020 (quarterly data)

	Farmland	Timberland	Commodities	Global Bonds	Global Stocks	Inflation
Farmland	1					

Timberland	0.4352	1				
Commodities	−0.1276	−0.1180	1			
Global Bonds	−0.0209	−0.0047	0.0006	1		
Global Stocks	0.1221	0.0215	0.4106	0.1465	1	
Inflation	−0.1724	0.0166	0.5404	−0.2180	0.0954	1

Sources: Global stocks, MSCI ACWI; global bonds, Bloomberg Barclays Global Aggregate Index; commodities, S&P GSCI Total Return; Farmland and Timberland, NCREIF

Overall, investments in natural resource assets exhibit significant portfolio diversification potential because of low correlations with traditional asset classes. Farmland and timberland display somewhat high correlation with each other (approximately 0.44), but each of these asset classes has slightly negative correlations with commodities.

Finally, commodity prices are often more sensitive to geopolitical factors and natural phenomena, such as weather conditions. Also, commodity investments frequently employ leverage and seek to capitalize on expected price changes based on more complex strategies, such as futures contract delivery timing and location, which will be addressed in later lessons.

Question Set

1. Contrast the pricing of commodities with the pricing of farmland/timberland investments.

Hide Solution

Solution:

Commodities are priced on public exchanges, and their pricing reflects the latest information. Land investments, such as farmland and timberland, are traded only when actual transactions occur; thus pricing is infrequent and relies on imprecise estimates.

2. Explain why commodity supply is usually slow to adjust to changes in demand for the commodity.

Hide Solution

Solution:

Commodity supply adjusts slowly to changes in commodity demand because producers are unable to alter supply quickly due to the need for extended lead times to affect production levels. For example, increased demand for a specific agricultural crop requires at least one growing cycle to produce more supply.

3. Explain one rationale as to why consumer price inflation is likely to be less volatile than commodity prices.

Hide Solution

Solution:

- One possible rationale is that consumer price inflation reflects many additional products beyond commodities (such as housing), which dampens the effect of commodity prices on inflation.
- Another possible rationale is that statistical techniques used to compute consumer price inflation cause smoothing in the data while commodity prices reflect real-time changes on public exchanges.

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Solution:

Commodities clearly perform well in periods of rising inflation and clearly exhibit negative performance as inflation declines. Global stocks and bonds do not exhibit such clear differences in performance related to inflation. Both global stocks and bonds perform best during periods of stable inflation.

OPTIONAL UNTESTED SEGMENT

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